Test Documentation — Vehicle Washing Management System

Cover Page

Table 1

Document identification data

Field	Value
Project	Vehicle Washing Management System (Vehicle Washing / LavaRápido)
Deliverable	Test documentation (section 2.5.4)
Program	Software Analysis and Development Technologist (ADSO)
Group (ficha)	3145556
Apprentices	Brayan Estiven Patiño Cabrera, Juan Camilo Guilombo Osorio, Merari Urbano Quintero
Instructor	Karol Daniela Correa
Training center	Industria
Date	October 20, 2026

This document gathers the test plan, the inventory of automated tests in the repositories, their actual results when run on October 19, 2026 (including the end-to-end, performance, and responsive design tests against the complete system), and the 42 functional test cases, run the same day, traced to the SRS v7.6 requirements (`lavarapido-docs/01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS`). On October 20, 2026, the inventory, case CP-42 and the traceability matrix were adjusted because the role change of an account was reverted in the code that day, and the wording of CP-10 because the customer cancels without writing a reason. Its structure follows the ISO/IEC/IEEE 29119-3 standard for test documentation (International Organization for Standardization [ISO], 2021). No result in this document is assumed.

1. Test Plan

1.1 Scope

The web application, the mobile application, and the seven backend services on their `develop` branch are tested. Offensive security testing is out of scope; load is only checked through the RNF-001 response time measurement.

1.2 Test Types and Tools

Table 2

Test types, actual tool, and where each applies

Type	What it verifies	Tool	Where
Domain unit tests	Rules of entities and value objects (states, validations)	JUnit 5 (`spring-boot-starter-test`); xUnit 2	`src/test/.../domain` in Java; `tests/PaymentService.Domain.UnitTests`
Application unit tests	Use cases with in-memory repositories (test doubles)	JUnit 5 with fakes (`Fakes.java`, `InMemory*Repository`); xUnit with `InMemoryPorts.cs`	`application/usecase` and `PaymentService.Application.UnitTests`
Web integration (controller + security)	Routes, HTTP status codes, JWT, and roles with MockMvc	`spring-boot-starter-webmvc-test`, `spring-security-test`	`*ApiWebTest.java`, `ApiSecurityWebTest.java`, `AdminNotificationControllerTest.java`
Gateway route integration	Each public route reaches its service	Spring Boot Test	`GatewayRoutesTest.java`
Persistence integration	EF Core mapping against a real SQL Server	xUnit + `DatabaseFixture`	`PaymentService.Infrastructure.IntegrationTests` (requires a database)
Web UI unit tests	Angular components, guards, and services	Vitest 4 + jsdom (`@angular/build:unit-test`)	`Web/src/**/*.spec.ts`

Type	What it verifies	Tool	Where
Functional / acceptance	End-to-end flows per role against the deployed system	Manual, with Postman or Swagger UI for the API and a browser / Expo Go for the UI	Cases CP-01 to CP-42 (section 4)
Cross-service (end-to-end) and RNF	Booking with operator choice and automatic assignment, complete flow of the three roles, inspection report, access rules per role; 95th percentile response time (RNF-001); web screens at 360 px (RNF-008)	Node 24 (`node:test` and `fetch`) and Playwright Core with headless Edge	`lavarapido-infra/e2e`

Note. Tools documented in JUnit (n.d.), xUnit.net (n.d.), and Vitest (n.d.).

1.3 Environment

Table 3

Execution environment for the tests (October 19, 2026)

Item	Version
Operating system	Windows 11 Home Single Language 10.0.26300
Node.js / npm	24.14.0 / 11.9.0
JDK	Java 25.0.4 (the projects compile for Java 21) with each service's Maven Wrapper
.NET SDK	9.0.306; the project targets `net8.0` and the tests were run with `DOTNET_ROLL_FORWARD=Major`
Database and messaging	SQL Server 2022, RabbitMQ 4.3, and Mailpit on Docker 28.4 (Docker Desktop). The unit tests do not need them; the 7 payment persistence tests used an isolated database, `LavaRapidoPaymentIT`, created with the same changelogs
Services	The six services and the gateway running on the machine (`./mvnw spring-boot:run` and `dotnet run`) against that infrastructure; email went to Mailpit so no real person was written to. The web with `ng serve` and the mobile app with `npx expo start --web`

1.4 Criteria

- **Passed:** the actual result matches the expected one.
- **Failed:** the result differs; the error is recorded.
- **Blocked:** could not run because a dependency was missing (for example, the database).
- **Exit:** the release is accepted when the automated suites pass and the high-priority functional cases are passed.

2. Automated Test Inventory

As of October 20, 2026, the repositories have 548 test methods: 505 in the backend and 43 in the web. The mobile app has no automated tests; it is checked with the TypeScript build and the screen walk-through (section 3.2).

Table 4

Tests per artifact (count of `@Test`/`@ParameterizedTest`, `[Fact]`/`[Theory]`/`[IntegrationFact]`, and `it()`)

Artifact	Files	Methods	What they cover
api-gateway	1	8	Routing of each `/api/v1/**` prefix to its service, and that `/internal/**` is not exposed by the gateway (`GatewayRoutesTest`)
security-service	17	133	Registration, sign-in with lockout after failed attempts, password recovery, profile and address, user administration (including the rule that an administrator account cannot be deactivated), roles and permissions, preferences, value objects, password policy, JWT issuing and HTTP security by role (`ApiSecurityWebTest`)
customer-service	6	50	Customer account, license plates, vehicle CRUD, `security.user_registered` listener and web API
booking-service	7	43	Booking life cycle, availability and the reason a time is not available, bays (`ServiceBayTest`), catalog, business data (`EstablishmentTest`), preferred operator in the event and web API

Artifact	Files	Methods	What they cover
operations-service	11	55	Manual and automatic assignment, free operators, start window, execution, rating, absences, operator metrics, team summary, operator profile, inspection report and web API
notification-service	12	106	Notification use cases, message factory per event, notices in the account's language, reminders, alerts for bookings without an operator, new version notice, device tokens and admin controller
payment-service (.NET)	9	110	Payment and receipt, reused reference, history with filters, promotions, loyalty points, payment policy, command and query services, and 7 persistence tests against SQL Server
Web (Angular)	37	43	Creation of components, guards and services (`should create` smoke tests), API error messages, and payment account selection
Mobile	0	0	No automated tests

> *Note.* Support files (`Fakes.java`, `InMemory*.java`, `InMemoryPorts.cs`, `DatabaseFixture.cs`) are not counted as tests. The 7 payment persistence tests skip themselves when the `PAYMENT_INTEGRATION_DB` variable is not set (`DatabaseFixture.cs:16-21`).

Backend tests are domain and application tests with in-memory doubles, plus controller tests with MockMvc. Most web tests check that each component is created; `api-error.spec.ts` and `payment.spec.ts` test real rules (the API error messages and the payment account selection).

3. Execution Results

On October 19, 2026, with the *develop* branch of the ten repositories updated that day, the eight automated suites were run: 655 tests, all passed and none skipped. This time the 7 payment persistence tests also ran, against the isolated `LavaRapidoPaymentIT` database. When the role change was reverted on October 20, 2 tests left security-service (`changingTheRoleReplacesItAndClosesTheSessions` and `nobodyChangesTheirOwnRole`), so that suite now has 2 tests fewer than the 151 in Table 5; Table 5 keeps the real numbers of the October 19 run.

Table 5

Actual results per artifact (October 19, 2026)

Artifact	Command	Run	Passed	Failed	Skipped	Duration
Web (Angular)	`npx ng test --watch=false`	43	43	0	0	6.7 s
api-gateway	`./mvnw test`	34	34	0	0	12.2 s
security-service	`./mvnw test`	151	151	0	0	9.7 s
customer-service	`./mvnw test`	69	69	0	0	7.4 s
booking-service	`./mvnw test`	43	43	0	0	6.8 s
operations-service	`./mvnw test`	55	55	0	0	7.6 s
notification-service	`./mvnw test`	112	112	0	0	0.6 s
payment-service: domain	`dotnet test PaymentService.sln`	113	113	0	0	0.3 s
payment-service: application	(same command)	28	28	0	0	0.4 s
payment-service: integration	(same command, with `PAYMENT_INTEGRATION_DB`)	7	7	0	0	2 s
Total		**655**	**655**	**0**	**0**	

> *Note.* Sources: Vitest output, the `./mvnw test` summary (`Tests run:`) and the surefire reports of each Java service, and the `dotnet test` output. Maven durations add up the time of the test classes, without the build. More tests run than there are methods in Table 4 because parameterized tests (`@ParameterizedTest`, `[Theory]`) run once per data item. In customer-service the `tests` attribute of the surefire XML says 68 because of a nested test class; Maven runs and counts 69, which is the right number (the previous version of this document had taken 68).

Figure 1

Result of the Java service tests (October 19, 2026)

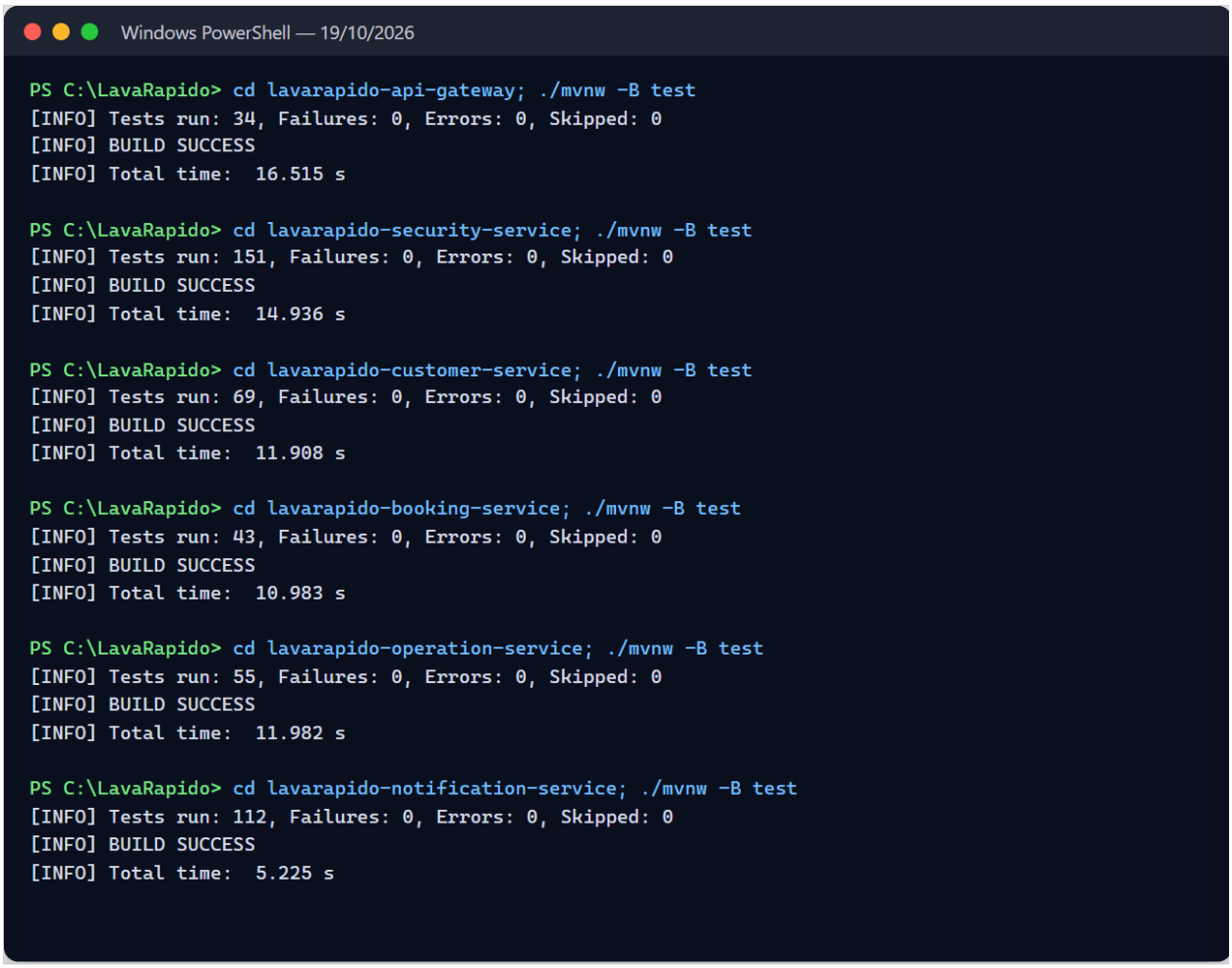


Figure 2

Result of the payment-service tests, including the persistence ones

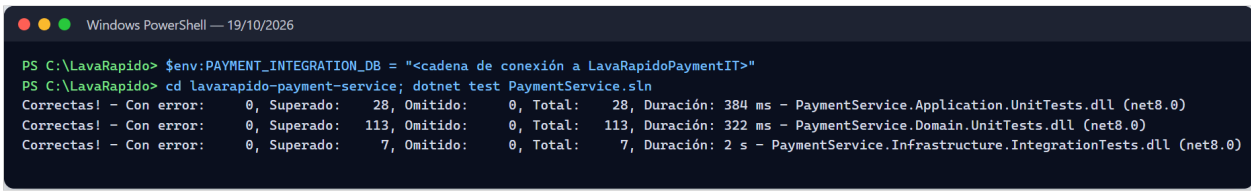
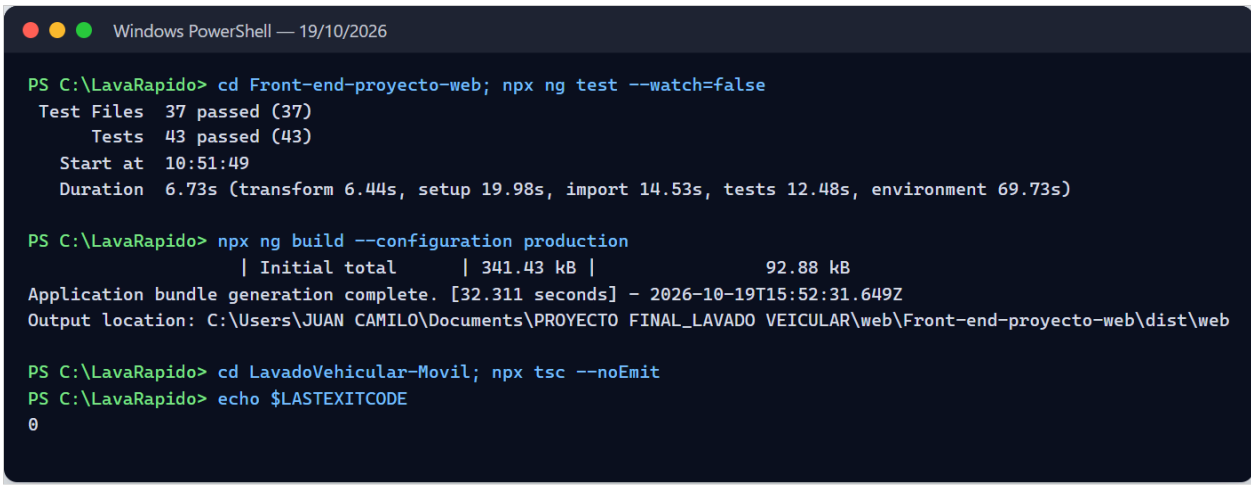


Figure 3

Web tests and production build, and TypeScript build of the mobile app



Execution remarks:

- The line `ERROR ... Could not send the password recovery email: SMTP down` in the security-service log is caused on purpose by `EmailResetCodeNotifierTest`, which simulates a down SMTP server. It is not a failure.
- Vitest shows a build warning: `src/test-providers.ts` is not in the `tsconfig`. It does not affect the results.
- The run used JDK 25 and .NET SDK 9 on projects for Java 21 and .NET 8. GitHub Actions runs the same tests with Temurin 21 and .NET 8 on every change (Deployment Document).

3.1 End-to-End Tests

On October 19, 2026, `node --test "e2e/*.test.js"` was run from `lavarapido-infra`, with the complete system running and `E2E_OPERATOR_ID=2` (the test operator of the local database). The 19 tests passed in 11.2 s. Each test deletes what it creates and leaves the customer's points as they were.

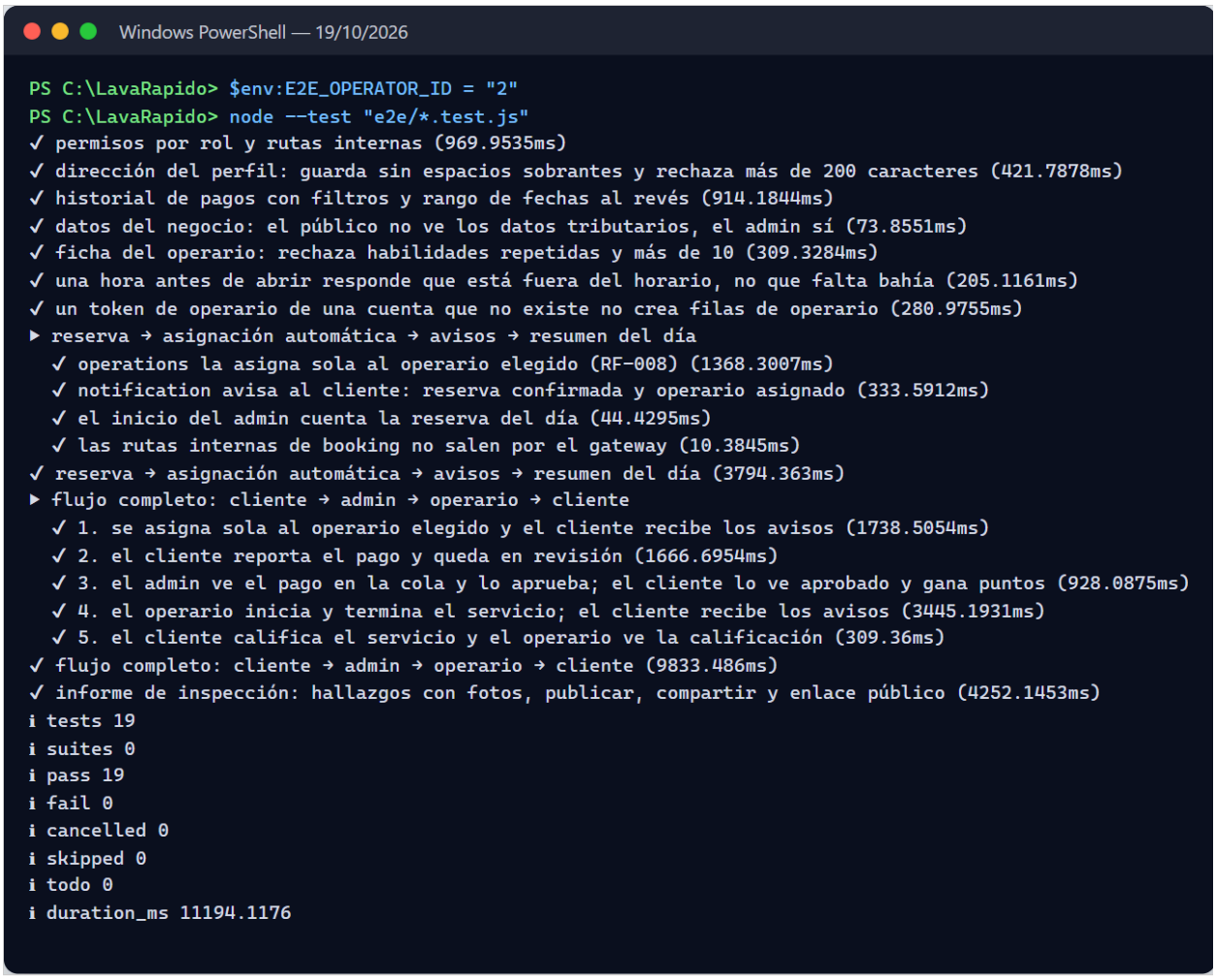
Table 5a

End-to-end tests (October 19, 2026)

File	Tests	What it walks through	Result
`access-rules.test.js`	7	Permissions by role and internal routes, profile address, payment history filters, tax data only for the administrator, operator profile, time outside opening hours (`reason: OUTSIDE_HOURS`), and the token of an account that does not exist	7 passed
`booking-flow.test.js`	5 (1 + 4 subtests)	Booking choosing an operator, automatic assignment through RabbitMQ, customer notices, day summary, and internal routes outside the gateway	5 passed
`full-flow.test.js`	6 (1 + 5 subtests)	Booking → reported payment → approval with points → service start and finish → rating, checking each notice	6 passed
`inspection-flow.test.js`	1	Inspection report with findings and photos, publishing, customer notice, and public link	1 passed
Total	**19**		**19 passed**

Figure 4

Output of the end-to-end tests



3.2 Performance and Responsive Design

The same day, the response time of the most used reads (NFR-001) was measured with `node e2e/performance.js`: 30 requests per endpoint through the gateway. All of them were far below the 2 s limit at the 95th percentile.

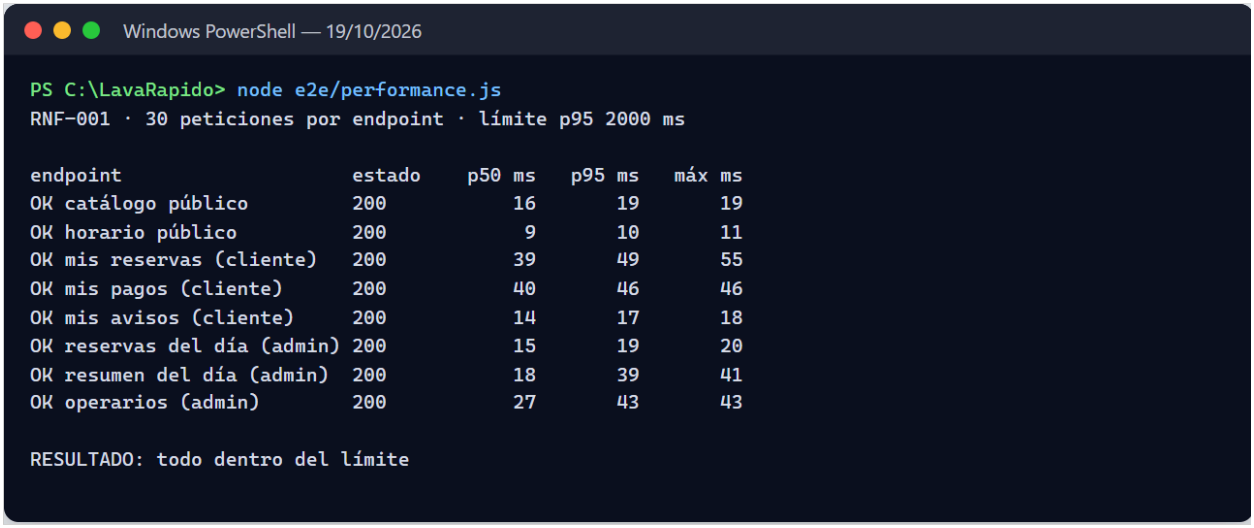
Table 5b

Response time per endpoint (NFR-001, October 19, 2026)

Endpoint	p50	p95	Maximum
Public catalog	16 ms	19 ms	19 ms
Public opening hours	9 ms	10 ms	11 ms
My bookings (customer)	39 ms	49 ms	55 ms
My payments (customer)	40 ms	46 ms	46 ms
My notices (customer)	14 ms	17 ms	18 ms
Day bookings (administrator)	15 ms	19 ms	20 ms
Day summary (administrator)	18 ms	39 ms	41 ms
Operators (administrator)	27 ms	43 ms	43 ms

Figure 5

Response time measurement



For responsive design (NFR-008), Playwright (headless Edge) went through the screens of the four web groups (public, customer, operator, and administrator) at 360 and 768 px, with the suite's `width-360.js` script and a check that also looks for elements past the right edge outside a container with its own scrolling.

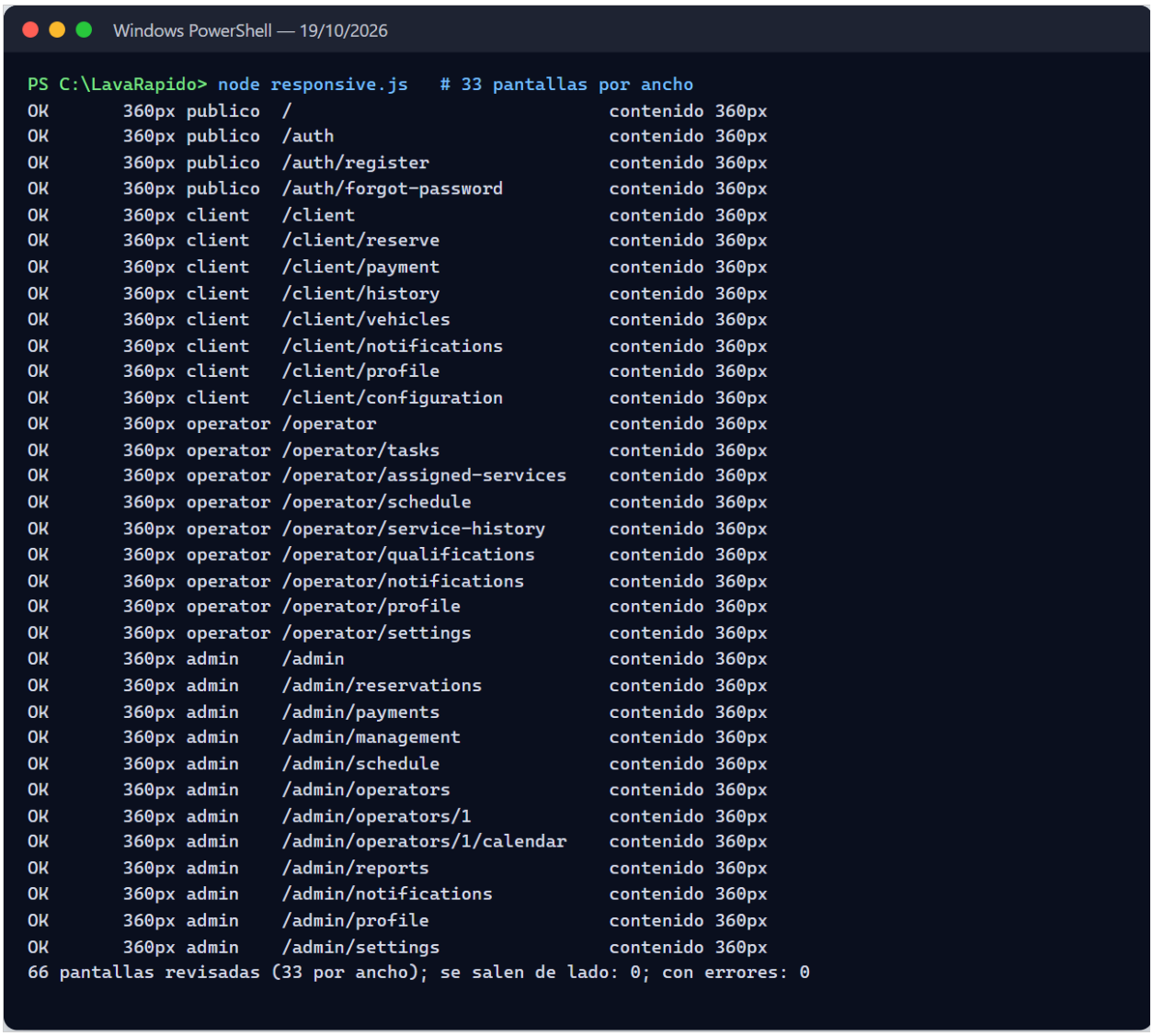
Table 5c

Responsive design check (October 19, 2026)

Test	Scope	Result
Web at 360 px (`e2e/width-360.js`)	12 screens: public, customer, and administrator	None goes beyond 360 px
Web at 360 px	33 screens: 4 public, 8 customer, 9 operator, and 12 administrator	None scrolls sideways or cuts content; no page errors or 5xx responses
Web at 768 px	The same 33 screens	None scrolls sideways or cuts content
Mobile app at 360 px (`npx expo start --web`)	15 screens: the five of the bottom menu of each role (customer, operator, and administrator), with temporary accounts deleted at the end	None scrolls sideways; no page errors
Build	`npx ng build --configuration production` (web) and `npx tsc --noEmit` (app)	No errors

Figure 6

Check of the web screens at 360 px



Note. Taking or picking photos with the phone camera and gallery cannot be automated in this test; it remains to be tested on an Android device.

4. Functional Test Cases

Cases CP-01 to CP-33 were defined on SRS v4 and first run on October 9, 2026; cases CP-34 to CP-42 cover the requirements implemented afterwards (RF-004b, RF-008, RF-010, RF-015, RF-022, RF-036, RF-037, and RF-038) and the administrator accounts of RF-028. All 42 were run on October 19, 2026, against the full system through the API Gateway (the same calls the web and mobile clients make), with a script that creates its own accounts and data and deletes them at the end. All 42 passed (Figure 7).

Table 6

Functional test cases

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-01	RF-001	Successful customer sign-up	Email not registered	Sign up and log in	ID, name, phone, new email, and valid password	Account created with the CLIENT role; the profile is created in customer-service	Sign-up 201; sign-in 200 with the CLIENT role; customer profile created in customer-service	Passed
CP-02	RF-001	Sign-up with a duplicate email	The email already exists	Sign up with the same email	Email from CP-01	"Email already registered" message; no account is created	409 `EMAIL_ALREADY_REGISTERED`; there is still one account with that email	Passed
CP-03	RF-002	Login by role	CLIENT, OPERATOR, and ADMIN users	Log in with each one	Valid credentials	Redirects to `/client`, `/operator`, or `/admin` by role	The three sign-ins return 200 with the CLIENT, OPERATOR, and ADMIN roles; the web redirects by that role	Passed
CP-04	RF-002	Wrong credentials	—	Log in with a wrong password, then with a nonexistent email	Two attempts	Same generic message in both; it does not reveal whether the email exists	Both: 401 `INVALID_CREDENTIALS` (same response)	Passed
CP-05	RF-003	Recover password	Registered user; Mailpit running	Request a code, read it in Mailpit, verify it, and set a new password	6-digit code	Password changed; login with the new one works	6-digit code read in Mailpit; verification 204; change 204; sign-in with the new password 200	Passed
CP-06	RF-003	Code locked after attempts	Code requested	Enter 5 wrong codes, then the right one	5 invalid codes	The right code is no longer accepted	The 5 wrong codes: 400; then the right one too: 400 `INVALID_RESET_CODE`	Passed
CP-07	RF-004	Password change with a wrong current password	Logged in	Change the password with the wrong current one	Wrong current password	"Incorrect current password" error; nothing changes	400 `INCORRECT_CURRENT_PASSWORD`; the previous password still works	Passed
CP-08	RF-005	Create a booking	Customer with an SUV	Choose vehicle, Basic service, a free date and time	SUV, Basic	Booking created; price COP 23,000 (seed price of Basic for SUV)	Booking created (201) with a total of $23,000, the seed price of Basic for SUV	Passed
CP-09	RF-006	Time slot taken	Active bays of a slot booked	Book the same slot until it is full	Several customer vehicles	The booking is not saved; up to five alternative times on the same day are offered	Three bookings took the three active bays; the fourth: 409 `SLOT_UNAVAILABLE` (`NO_FREE_BAY`) with 5 alternative times	Passed
CP-10	RF-007	Cancel a booking before it starts	Booking CONFIRMED	Cancel as the customer (the customer writes no reason)	—	Booking is CANCELLED with the reason `CUSTOMER_REQUEST` set by the system	200, booking CANCELLED with the reason `CUSTOMER_REQUEST` stored	Passed
CP-11	RF-007	Change a booking in progress	Booking IN_PROGRESS	Try to cancel	—	Action blocked with a message	409 `BOOKING_NOT_CHANGEABLE`	Passed
CP-12	RF-009	Assign an unavailable operator	Operator with an absence at the booking time	Check candidates and try to assign them; then assign an available one	Absent operator	The operator is not allowed; available operators are listed first	The operator without a shift shows `OPERATOR_NOT_ON_SHIFT` and assigning them returns 409; the available ones come first; assigning the available one: 200	Passed
CP-13	RF-011	Operator sees their services	Booking assigned to the operator	Check assigned services	—	List with customer and vehicle data	200; the service carries the customer name, plate, vehicle, and services	Passed
CP-14	RF-012b	Start and finish a service	Assigned service PENDING, 15 min before its time	Another operator tries to start; the assigned one starts and finishes	—	States IN_PROGRESS and COMPLETED; the other operator gets an error	Another operator: 404 `BOOKING_NOT_FOUND`; the assigned one starts (200) and finishes (200); the booking ends COMPLETED	Passed

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-15	RF-012	Rate a service once	Booking COMPLETED	Rate 5 stars; try to rate again	5, comment	First rating saved; the second is rejected	First rating 201; the second 409 `ALREADY_RATED`	Passed
CP-16	RF-013	Payment methods	Payment accounts configured	Check methods (admin) and accounts (customer)	—	Cash, Nequi, Daviplata, and bank transfer; cash shows no QR	Methods: Cash, Nequi, Daviplata, and Transfer; the customer sees the active accounts; cash without QR	Passed
CP-17	RF-013b	Upload a receipt	Booking awaiting payment	Pay with Nequi attaching image, amount, and reference	PNG < 10 MB	Payment created in review with its receipt	201; payment IN_REVIEW with its receipt and the amount computed by the server	Passed
CP-18	RF-014	Approve and reject payments	Payments in review	Approve one and reject another with a reason	Rejection reason	States APPROVED and REJECTED; the customer gets a notification	APPROVED and REJECTED; the customer got PAYMENT_CONFIRMED and PAYMENT_REJECTED	Passed
CP-19	RF-016	Duplicate reference	Approved payment with reference R1	Upload another receipt with R1 on another booking	Reference R1	Receipt rejected	409 `TRANSACTION_REFERENCE_REUSED`	Passed
CP-20	RF-018	Booking notification	`MESSAGING_ENABLED=true`	Confirm a booking; open notifications; mark as read	—	The notification arrives; the unread counter goes down	BOOKING_CONFIRMED arrived; the unread counter dropped by 1 when it was marked	Passed
CP-21	RF-020	Language saved on the account	Logged in	Switch to English; start a new session	`en`	The UI opens in English	Saved 200; in a new session the account language is English	Passed
CP-22	RF-021	Schedule exception	—	Create a "closed" exception and check availability that day	10/20/2026	No times available that day	Exception created (201); there are no free times that day	Passed
CP-23	RF-024	Create and deactivate a promotion	Admin session	Create a 10% promotion and deactivate it	10% discount	Promotion visible, then inactive	Created (201) and deactivated (200); the list shows it paused	Passed
CP-24	RF-026	Earn points	Service with 10 points configured	Approve the booking payment; check the balance	—	The balance grows by the service points	The balance goes up 30 points, those of the service	Passed
CP-25	RF-027	Deactivate one's own account	Customer session	Deactivate the account with the current password and try to log in	Current password	The session is revoked and login is rejected	Deactivation 204; the new sign-in returns 403 `ACCOUNT_DISABLED`	Passed
CP-26	RF-028	Create an operator and restrict by role	Admin and customer sessions	Admin creates an operator; the customer calls `GET /admin/users`	Operator data	OPERATOR account created; the customer gets 403	OPERATOR account created (201) and listed; the customer gets 403	Passed
CP-27	RF-029	Edit the permission matrix	Admin session	Remove `EDIT_DATA` from OPERATOR; try to create a role	Permission `EDIT_DATA`	The matrix is saved; roles cannot be created	Matrix saved (200); creating a role returns 405 (roles are fixed)	Passed
CP-28	RF-030	Register a vehicle with a duplicate plate	Customer with a registered vehicle	Register another vehicle with the same plate	Plate FPW653	Registration is rejected	409 `PLATE_ALREADY_REGISTERED`	Passed

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-29	RF-031	Price change without affecting bookings	Confirmed booking with Basic	Admin raises the Basic/SUV price; check the existing booking and create a new one	COP 23,000 → 24,000	The existing one keeps its price; the new one uses the new price	The existing booking keeps $23,000; the new one costs $24,000	Passed
CP-30	RF-032	Bay under maintenance	BAY-04 under maintenance (seed)	Fill a slot and check which bays were used	—	The bay under maintenance is not used for new bookings	With Bay 1 under maintenance, two bookings at the same time went to bays 2 and 3 and the third got 409	Passed
CP-31	RF-033	Disallowed status transition	Booking CANCELLED	Admin tries to change it to CONFIRMED	—	Rejected	409 `INVALID_STATUS_TRANSITION`	Passed
CP-32	RF-034	Deactivated payment account	Active Daviplata account	Admin deactivates it; the customer checks accounts	—	The account is no longer shown to the customer	Account deactivated (200); the customer no longer sees it	Passed
CP-33	RF-035	Refund an approved payment	Payment APPROVED	Admin refunds	—	Status REFUNDED; `payment.refunded` is published and the customer is notified	Status REFUNDED; the customer got PAYMENT_REFUNDED and the booking points were reversed	Passed
CP-34	RF-004b	Lockout after failed attempts	Active account	Sign in 5 times with a wrong password and then with the right one	Email of a new account	At the fifth failure the account is locked for 15 minutes; the right password does not work either	Attempts: 401, 401, 401, 401, and 423; with the right password: 423 `ACCOUNT_LOCKED`	Passed
CP-35	RF-008	Automatic operator assignment	Operators with a shift that day	Book without choosing an operator	Basic service, SUV	The booking is assigned to a free operator without the administrator	The booking was assigned to a free operator without the administrator	Passed
CP-36	RF-010	The customer chooses the operator	The operator has a shift and is free	List the free operators and book choosing one	`preferredOperatorId`	The booking stays with the chosen operator	The chosen operator shows as free and the booking stayed with them	Passed
CP-37	RF-015	Payment history with filters	The customer has two payments	Query the history with the day of the paid bookings, with another day, and with an inverted range	Dates	Only that day's payments come out; the inverted range is rejected	With the day of the paid bookings, their 2 payments come out; with another day, none; the inverted range returns 400 `INVALID_DATE_RANGE`	Passed
CP-38	RF-022	Operator absence	Operator with a shift	Record an absence that day, list the candidates for a booking that day, and record a past absence	Whole-day absence	The operator shows as absent; the past absence is rejected	Absence recorded (201); that day the operator shows `OPERATOR_ABSENT`; the past absence returns 400 `ABSENCE_IN_PAST`	Passed
CP-39	RF-037	Business data to be completed	System with migration 111 applied	Read the data, save an invalid phone, and then a valid support line	3-digit phone	The real business data is empty; the invalid phone is rejected; the valid change shows on the public page	The tax data and channels were empty; the invalid phone returns 400 `ESTABLISHMENT_INVALID_PHONE`; the valid change (200) shows on the public page	Passed
CP-40	RF-038	Operator profile	Operator created	Save a specialty and a skill	EXPERT skill	The profile is saved	200; profile saved with its specialty and one skill	Passed

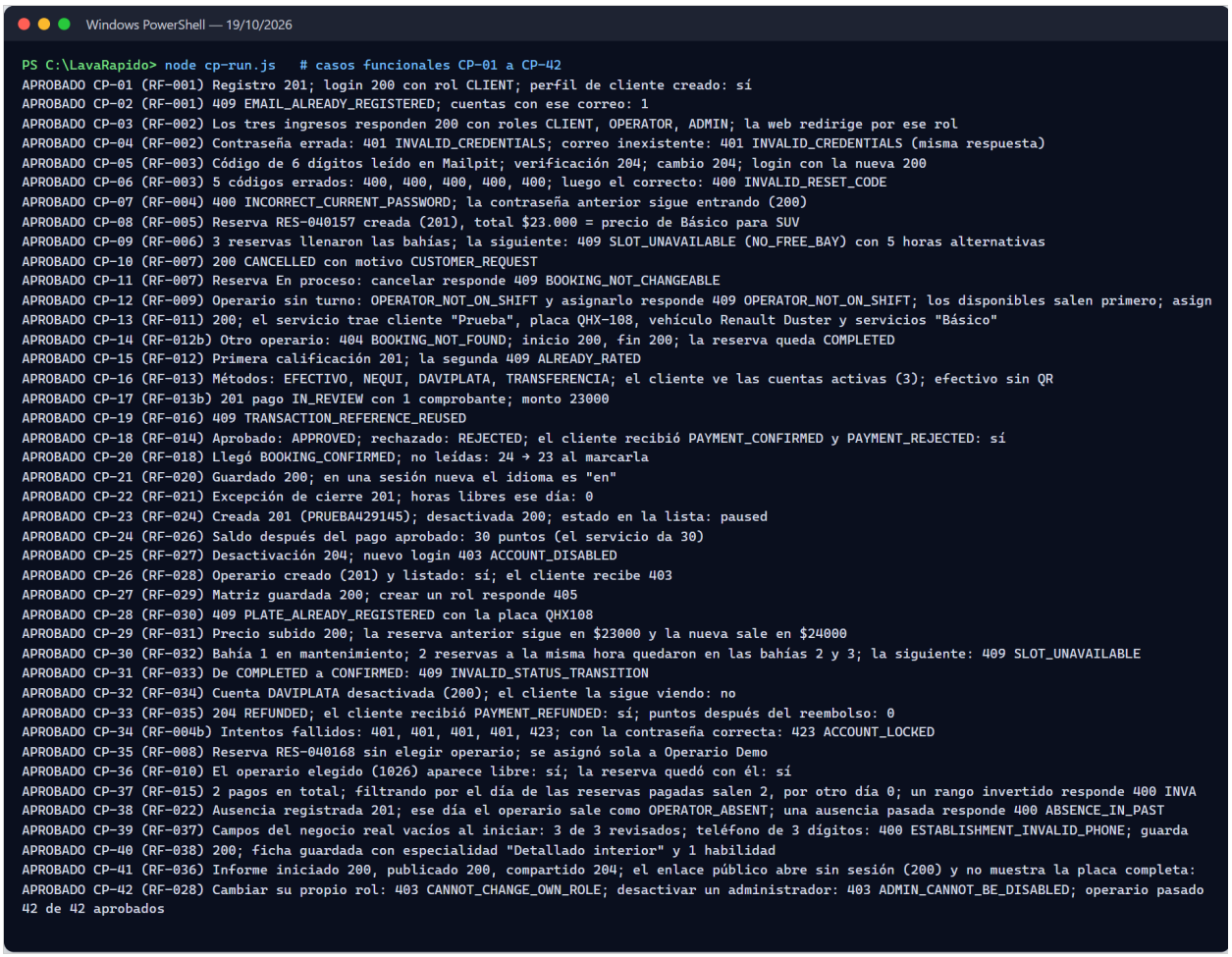
ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-41	RF-036	Inspection report	Confirmed booking of a 200-minute service	Start the report, finish a phase, publish, share, and open the public link	Temporary 200-minute service	The customer gets the notice and the link opens without a session with the plate masked	Report started (200), published (200), and shared (204); the customer got INSPECTION_REPORT and the link opens without a session and without the full plate	Passed
CP-42	RF-028	An administrator account cannot be deactivated	Administrator session	Try to deactivate an administrator account	—	Rejected: an administrator account cannot be deactivated	403 `ADMIN_CANNOT_BE_DISABLED`	Passed

> *Note.* On October 19, CP-42 also tested the role change of an account (403 `CANNOT_CHANGE_OWN_ROLE` when changing one's own role, and an operator turned into a customer with 200). That part was removed from the case on October 20, 2026, when the feature was reverted in the backend, the web and the app.

> *Note.* The prices in the test data come from the seed changelogs (`103-seed-catalog.sql`, `104-seed-booking-catalogs.sql`). The script creates a test customer, two operators, and an administrator, with emails of the `prueba.lavarapido.test` domain, and its own vehicles and bookings. At the end it deletes everything it created and restores the Basic price, the permission matrix, the bay, the payment account, and the business data; the database was checked to have nothing left. The recovery emails were read in Mailpit. Evidence: execution log with the HTTP code and the error code of each step (Figure 7).

Figure 7

Run of the 42 functional cases (October 19, 2026)



5. Traceability Matrix

Of the 41 requirements in SRS v7.6, 38 have a functional case and 39 a related automated test; RF-032 and RF-034 only have a functional case, and RF-017, RF-023, and RF-025 only an automated test.

Table 7

RF → status → endpoint / screen → test case

RF	Status (SRS v7)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related automated test
RF-001	Implemented	`POST /auth/register`	`/auth/register`	CP-01, CP-02	`AccountRegistrationServiceTest`, `UserRegisteredListenerTest`
RF-002	Implemented	`POST /auth/login`	`/auth`	CP-03, CP-04	`AuthenticationServiceTest`, `JwtTokenIssuerTest`, `ApiSecurityWebTest`
RF-003	Implemented	`POST /auth/password/forgot`, `/verify`, `/reset`	`/auth/forgot-password`	CP-05, CP-06	`PasswordRecoveryServiceTest`, `PasswordResetTokenTest`
RF-004	Implemented	`PUT /users/me/password`	`/client/profile`	CP-07	`UserProfileAndAdministrationTest`, `PasswordPolicyTest`
RF-004b	Implemented	`POST /auth/login` (423 `ACCOUNT_LOCKED`)	`/auth`	CP-34	`AuthenticationServiceTest` (lockout and window), `PasswordRecoveryServiceTest` (a reset lifts it), `ApiSecurityWebTest` (423)
RF-027	Implemented	`GET/PATCH /users/me`, `GET/PUT /users/me/address`, `PUT /users/me/email`, `POST /auth/logout`, `POST /users/me/deactivate`	`/client/profile`	CP-25	`UserProfileAndAdministrationTest`, `PersonAddressServiceTest`, `AuthenticationServiceTest`

RF	Status (SRS v7)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related automated test
RF-028	Implemented	`GET/POST /admin/users`, `PATCH /admin/users/{id}/status`	`/admin/management`, `/admin/operators`	CP-26, CP-42	`UserProfileAndAdministrationTest`, `ApiSecurityWebTest`
RF-029	Implemented	`GET /admin/roles-permissions`, `PUT /admin/roles-permissions/{role}`	`/admin/management`	CP-27	`RolePermissionAdministrationServiceTest`
RF-030	Implemented	`/vehicles`, `GET /vehicle-types`, `GET /admin/vehicles`	`/client/vehicles`	CP-28	`CustomerVehicleServiceTest`, `LicensePlateTest`, `CustomerApiWebTest`
RF-031	Implemented	`GET /catalog/services`, `/admin/catalog/services`	`/admin/management`	CP-29	`CatalogServiceTest`
RF-021	Implemented	`PUT /admin/schedule/business-hours`, `POST /admin/schedule/exceptions`	`/admin/schedule`	CP-22	`AvailabilityPolicyTest`
RF-032	Implemented	`/admin/bays`	`/admin/schedule`	CP-30	—
RF-037	Implemented	`GET/PUT /admin/establishment`, `GET /establishment`	`/admin/settings`	CP-39	`EstablishmentTest`, `BookingApiWebTest`
RF-005	Implemented	`POST /bookings`, `GET /catalog/services`	`/client/reserve`	CP-08	`BookingServiceTest`, `BookingTest`
RF-006	Implemented	`GET /bookings/availability`	`/client/reserve`	CP-09	`AvailabilityPolicyTest`
RF-007	Implemented	`PUT /bookings/{id}`, `POST /bookings/{id}/cancel`	`/client/history`	CP-10, CP-11	`BookingTest`, `BookingApiWebTest`
RF-033	Implemented	`GET /admin/bookings`, `PATCH /admin/bookings/{id}/status`	`/admin/reservations`	CP-31	`BookingTest`, `BookingServiceTest`
RF-008	Implemented	`booking.confirmed` event, `GET /internal/v1/bookings`	`/admin/reservations`	CP-35	`AutoAssignmentPolicyTest`, `OperationsServiceTest` (automatic assignment, deactivated account), `booking-flow.test.js`
RF-009	Implemented	`GET /admin/assignments/{id}/candidates`, `PUT /admin/assignments/{id}`	`/admin/reservations`	CP-12	`AssignmentPolicyTest`, `OperationsServiceTest`
RF-010	Implemented	`GET /operators/available`, `POST /bookings` (`preferredOperatorId`)	`/client/reserve`	CP-36	`OperationsServiceTest` (free operators), `OperationsApiWebTest`, `booking-flow.test.js`
RF-011	Implemented	`GET /operator/services`	`/operator/assigned-services`	CP-13	`OperationsServiceTest`
RF-012b	Implemented	`POST /operator/services/{id}/start`, `/finish`	`/operator/assigned-services`	CP-14	`StartPolicyTest`, `ServiceExecutionTest`
RF-012	Implemented	`POST /ratings/{bookingId}`	`/client/history`	CP-15	`ServiceExecutionTest`
RF-022	Implemented	`PUT /admin/operators/{id}/availability`, `/absences`, `GET /admin/operators/summary`	`/admin/operators/:id`	CP-38	`AssignmentPolicyTest`, `TeamSummaryCalculatorTest`
RF-036	Implemented	`/admin/inspections/{bookingId}/**`, `GET /public/inspections/{token}`	`/admin/reservations/:id/inspection`, `/report/:token`	CP-41	`InspectionReportTest`, `OperationsApiWebTest`
RF-038	Implemented	`GET /admin/operators/profiles`, `PUT /admin/operators/{id}/profile`	`/admin/operators/:id`	CP-40	`OperatorProfileServiceTest`
RF-034	Implemented	`/admin/payment-accounts`, `GET /payment-accounts`	`/admin/settings`	CP-32	—
RF-013	Implemented	`GET /payment-accounts`	`/client/payment`	CP-16	`PaymentQueryServiceTests`
RF-013b	Implemented	`POST /payments`	`/client/payment`	CP-17	`PaymentTests`, `PaymentCommandServiceTests`
RF-014	Implemented	`POST /admin/payments/{id}/approve`, `/reject`	`/admin/payments`	CP-18	`PaymentTests`, `PaymentCommandServiceTests`
RF-015	Implemented	`GET /payments/me?from=&to=&service=`	`/client/history`	CP-37	`PaymentQueryServiceTests` (order and filters)
RF-016	Implemented	`POST /payments`	`/client/payment`	CP-19	`PaymentPolicyTests`, `PaymentCommandServiceTests` (reused reference)
RF-035	Implemented	`POST /admin/payments/{id}/refund`	`/admin/payments`	CP-33	`PaymentTests`, `PaymentCommandServiceTests`

RF	Status (SRS v7)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related automated test
RF-017	Implemented	`GET /admin/bookings/day-summary`, `GET /operators/my-bookings` and dashboard stores	`/client`, `/operator`, `/admin`	—	`BookingServiceTest` (day summary), web smoke tests (`dashboard.spec.ts`)
RF-018	Implemented	`/notifications/*` + RabbitMQ events	`/*/notifications`	CP-20	`NotificationUseCasesTest`, `NotificationFactoryTest`, `BookingReminderServiceTest`, `LocalizedNotificationsTest`, `UnassignedBookingAlertTest`, `StaffNotificationsTest`
RF-019	Implemented	`PUT /users/me/preferences`	`/*/configuration`	Covered by CP-21	`UserPreferenceServiceTest`
RF-020	Implemented	`PUT /users/me/preferences`	`/*/configuration`	CP-21	`UserPreferenceServiceTest`, `UserPreferenceTest`
RF-023	Partial	`GET /admin/operators/{id}/metrics`, `GET /admin/operators/summary`, payment and booking stores	`/admin/reports`, `/admin/operators/:id`	—	`OperatorMetricsCalculatorTest`, `TeamSummaryCalculatorTest`
RF-024	Implemented	`/admin/promotions`	`/admin/management`	CP-23	`PromotionTests`
RF-025	Implemented	`POST /loyalty/redeem`	`/client/payment`	—	`CustomerPromotionServiceTests`
RF-026	Implemented	`GET /loyalty/balance`	`/client`	CP-24	`LoyaltyTransactionTests`

Note. The automated test column points to the file whose name and methods match the requirement; code coverage was not measured. `booking-flow.test.js` is in `lavarapido-infra/e2e` and needs the system running.

6. Findings and Open Items

Table 8

Testing gaps found

#	Gap	Recommendation
1	The mobile app has no automated tests	Add Jest with `jest-expo`, at least for validators (`shared/validators`) and services (`core/services`)
2	Most of the 43 web tests only check that the component is created	Test more real rules: `authGuard` by role, `authInterceptor` on 401, booking form
3	The 7 payment persistence tests need a migrated database and skip without it; on October 19 they were run against `LavaRapidoPaymentIT` and passed	Document in the README how to create the test database and set `PAYMENT_INTEGRATION_DB`, and run them in GitHub Actions with a SQL Server service
4	For RF-023 (reports) only the operator metrics and the team summary are tested; the PDF/Excel export has no test	Add a functional case for reports and the PDF/Excel export
5	RF-032 (bays) and RF-034 (payment accounts) have no dedicated automated tests: `ScheduleService` and `PaymentAccountService` have no test class	Add unit tests for those two services
6	The requirements implemented after October 9 (RF-004b, RF-008, RF-010, RF-015, RF-022, RF-036, RF-037, and RF-038) had no functional case of their own	Closed on October 19, 2026, with cases CP-34 to CP-41

Table 9

Defects found while running the system

#	Defect	Evidence	Severity
1	The administrator dashboard always says "¡Hola, Laura!" regardless of who logs in	`Web/src/app/features/admin/pages/dashboard/dashboard.ts:73` (`adminName = 'Laura'`); "Laura Méndez" is also hard-coded in `payment-review-modal.html:167`	Low

#	Defect	Evidence	Severity
2	In the operator's "Assigned services" (web), the Customer column shows the license plate	`Web/src/app/features/operator/pages/assigned-services/assigned-services.ts:141`	Medium
3	On the customer home (mobile), "Assigned operator" is empty even though the booking has an operator	Booking RES-060003, assigned in CP-12; mobile screenshot	Medium
4	The brand name is inconsistent: "AquaWash" (mobile), "Car Wash" and "Lavado Vehicular" (web), "Express Car Wash" (establishment)	Web and mobile screenshots	Low
5	The landing page advertises a "mobile wash service" and "wherever you are", but SRS v4 defines a single-location car wash	Landing screenshot; SRS v4, section 1	Low
6	If there are two accounts of the same method (two Nequi), the payment screen shows both selected at once	Customer Payments screenshot	Low
7	When logging out, the web and the app showed "Your session expired", because a 401 that arrived after the token was deleted was taken as an expired session	`auth-interceptor.ts` (web) and `AuthServiceContext.tsx` (mobile)	Medium
8	On a phone (360 px) there were buttons off the screen in the landing bar, overlapping titles, the ☰ button over the titles, Assigned services badges off the screen, Management filters outside their card, and Settings switches over their text	Responsive design check of October 17, 2026 (Table 5c)	Medium
9	In the mobile app, the date and search inputs were squashed when placed in a column (History and Ratings)	`shared/components/screen/FilterInputs.tsx`	Low
10	The customer's next service progress showed 10% or 25% although the wash had not started	`features/client/pages/dashboard/dashboard.ts` (web) and `modules/client/models/client.ts` (mobile)	Low

Note. Defects 7 to 10 were fixed between October 16 and 17, 2026, on the *develop* branch of the web and the app; 8 and 9 were measured again with no overflow (Table 5c).

Note. Code review of October 16, 2026: the six defects are fixed. The dashboard greets with the session name and the payment review shows the reviewer (1); the Client column shows the customer name (2); the customer home shows the assigned operator on the web and on mobile (3); the brand is LavaRápido on the web and on mobile (4); the home page describes the service at the car wash (5); and the payment screen selects by account and shows the holder when two accounts share a method (6, `payment.spec.ts`).

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